

## Exercise 1: Frequency-Based Vectorization Techniques in NLP

In this exercise, you will apply different frequency-based vectorization techniques commonly used in Natural Language Processing (NLP). You are given a small corpus consisting of four documents.

### Given Corpus

- D1: cats like milk
- D2: dogs like bones
- D3: cats and dogs play
- D4: milk and bones are food

### Tasks

Using the given corpus, build the following document-term matrices manually.

#### 1. Bag of Words (BoW)

Construct the Document-Term Matrix using the Bag of Words approach.

- Identify the vocabulary of the corpus.
- Create the document-term matrix.
- Use raw term frequencies.

#### 2. Bag of N-grams (n = 2)

Construct the Document-Term Matrix using bigrams (2-grams).

- Extract all unique bigrams from the corpus.
- Build the document-bigram matrix.
- Use raw frequencies.

#### 3. TF-IDF Vectorization

Construct the TF-IDF matrix for the corpus.

- Compute the Term Frequency (TF) for each term.
- Compute the Inverse Document Frequency (IDF).
- Compute the TF-IDF weights.
- Present the final TF-IDF matrix.

## Exercise 2: Preparing Training Data for Word2Vec

- Using the same corpus from exercise 1, prepare the training samples required for training Word2Vec models using both Skip-gram and CBOW architectures.
- Assume a context window size of 1.
- For the Skip-gram model, generate all training pairs.
- For the CBOW model, generate all training pairs.